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Structural, Optical, Electrical and Electronic Properties of PEDOT: PSS Thin Films and Their Application in Solar Cells

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Abstract
The structural, optical, electrical and electronic properties of poly(3,4-ethylene dioxythiophene):polystyrene sulfonate (PEDOT:PSS) thin films are investigated using transmittance spectrometry, atomic force microscopy, water droplet contact angle measurement, Raman scattering spectrometry, and photoelectron emission spectrometry. The fabrication parameters of the thin films were varied in order to determine which would improve the performance of the resultant PEDOT:PSS thin-film-based transparent conductive electrodes (TCEs) and hole-transporting materials (HTMs) in solar cells. For TCE applications, the experimental results lead to the conclusion that the necessary transparency and electrical conductivity of PEDOT:PSS (1:2.5 wt%) thin films can be simultaneously obtained by using a post-immersion treatment process which encourages the formation of connected conductive PEDOT channels and an increased carrier density in the PEDOT chains. On the other hand, the work function of the HTMs depends on the molecular structure of the PEDOT chains in the PEDOT:PSS thin films, which can be manipulated by varying the PEDOT/PSS ratio, spin rate, thermal annealing conditions and the viscosity of the PEDOT:PSS solutions. The work function

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of PEDOT:PSS thin films increases from 5.0 eV to 5.4 eV with an increase in the linearly oriented PEDOT chains, which results in an improvement in the open-circuit voltage of the solar cells.

Keywords: PEDOT:PSS thin films, molecular structure, free carriers, work function, viscosity

8.1 Introduction

Poly(3,4-ethylenedioxythiophene):polystyrene sulfonate (PEDOT:PSS) thin films were first used as the hole-injection contact in light-emitting diodes [1–4] and electrochemical devices [5–8] because PEDOT is a stable conductive polymer [9]. Owing to their high electrical conductivity [10–12] and large work function [13–15], the PEDOT:PSS thin films can be used as transparent conductive electrodes (TCEs) [16–18] or hole-transporting materials (HTMs) [19–21] in solar cells. PEDOT is a π-conjugated conductive polymer which is electrically attracted to an insulator-like PSS segment and therefore can be dissolved in aqueous solutions, which is beneficial in terms of the fabrication cost because PEDOT:PSS thin films can be easily obtained using solution processes. PEDOT:PSS (1:2.5 wt%) aqueous solutions are widely used as the starting material for the fabrication of transparent conductive electrodes (TCEs). The electrical conductivity is usually smaller than 1 S/cm when a PEDOT:PSS (1:2.5 wt%) thin film is deposited on top of a glass substrate using the spin-coating method due to the poor connection between the PEDOT chains in the PEDOT:PSS thin films. The electrical conductivity of PEDOT:PSS (1:2.5 wt%) thin films can be dramatically increased from ~1 S/cm to ~1000 S/cm using various treatment methods [10, 17, 18]. These modified highly conductive PEDOT:PSS (1:2.5 wt%) thin films can be used as stand-alone TCEs for ITO substitutes. Previous reports [22, 23] have explained the increased electrical conductivity in the modified PEDOT:PSS (1:2.5 wt%) thin films as being due to the partial removal of the insulator-like PSS segments and the formation of conductive PEDOT three-dimensional (3D) networks [24]. The starting material for the HTMs is a PEDOT:PSS (1:6 wt%) aqueous solution. The spin coating of a p-type PEDOT:PSS thin film on top of an n-type ITO/glass sample can lead to an improvement in the open-circuit voltage (V_{OC}) of the solar cells, and block the electron recombination from the conduction band in the active layer to the Fermi level in the ITO thin film.

The transparency, electrical conductivity and work function of PEDOT:PSS thin films is greatly dependent on their molecular structure which consists of amorphous PSS segments, PEDOT-PSS core-shell clusters and linear PEDOT islands [24]. Under air-dried conditions, there is no clear segregation between the amorphous and crystalline phases, which indicates that the PEDOT chains are randomly distributed in the PEDOT:PSS thin films. After annealing in air at 120–160 °C for 10–20 min, uniformly distributed PEDOT-PSS core-shell clusters form with an average diameter of ~40 nm, which is close to the film thickness. The PEDOT-PSS core-shell clusters are separated by amorphous PSS polymers, which impede carrier transports in the transverse direction. The core-shell clusters are hole-transporting pathways, which means that the work function of the PEDOT:PSS thin films is dominated by the molecular structure in the PEDOT-PSS core-shell clusters. This PEDOT molecular structure can be manipulated by varying the PEDOT/PSS ratio [14, 25], spin rate [15, 26], thermal annealing conditions [26, 27] and viscosity of the PEDOT:PSS solutions [28].

This chapter will explore the correlation between the properties of the PEDOT:PSS thin films and the photovoltaic performance of the resultant solar cells. The molecular structure of the thin films is characterized by examining the Raman scattering spectra. The free carrier density of the PEDOT chains in the PEDOT:PSS thin films can be determined by fitting the transmittance spectrum with a transfer matrix method [29] and Drude model [30]. The work function of the PEDOT:PSS thin films is measured by a photoelectron emission spectrometer. The hydrophilicity and surface roughness of the PEDOT:PSS thin films are assessed using a homemade contact-angle imaging system and an atomic force microscope (AFM), respectively.

8.2 Chemical Structure of PEDOT:PSS

The chemical structure of PEDOT:PSS is depicted in Figure 8.1a. The positively charged PEDOT chains and negatively charged PSS chains are attracted through Coulomb interaction. The molecular structure of the PEDOT chains can be a benzoid form and or a quinoid form, as shown in Figure 8.1b. The benzoid form has a coil conformation, while the quinoid form has a linear or expanded-coil conformation. Both the benzoid form (coil conformation) and quinoid form (linear or expanded-coil conformation) can coexist in a PEDOT:PSS thin film. Theoretical studies reveal that the PEDOT chain is bent towards the PSS moiety and, thus, PEDOT:PSS is likely to form a partially coiled or helical structure [31]. Figure 8.1c presents the Raman fingerprints of a PEDOT:PSS thin film, which can be used to investigate the chemical structure of PEDOT:PSS thin films [32–34].

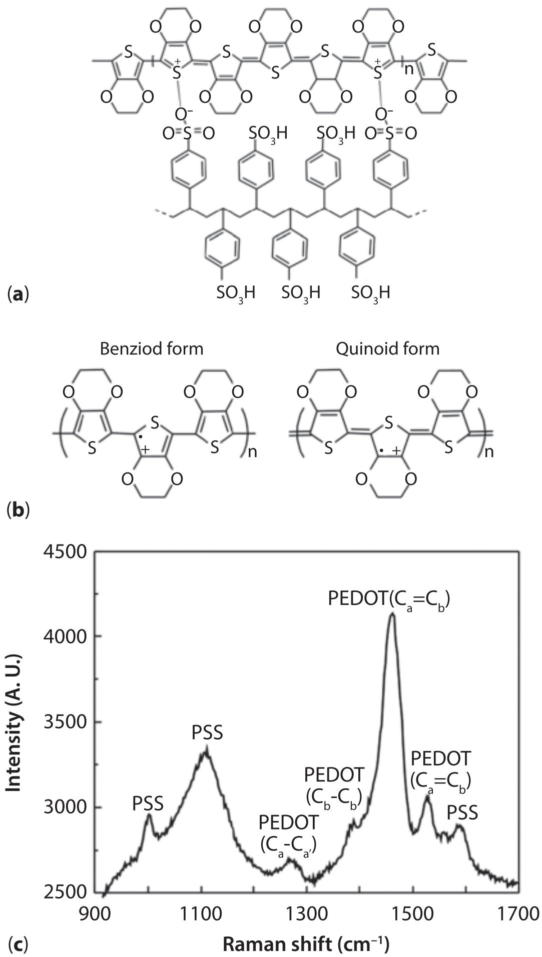

Figure 8.1 (a) Molecular structures of PEDOT and PSS polymers. (b) Benzoid and quinoid forms of PEDOT polymers. (c) Raman fingerprint of a PEDOT:PSS (1:6 wt%) thin film.

The vibrational modes of the PSS segments are located at 1110 cm^{-1} and 1000 cm^{-1}, while the vibrational modes of the PEDOT chains are located at 1524 cm^{-1}, 1452 cm^{-1}, 1383 cm^{-1}, and 1272 cm^{-1}, which have been assigned to the $C_a = C_b$ asymmetrical, $C_a = C_b$ symmetrical, C_b-C_b stretching, and C_a-$C_{a'}$ inter-ring stretching vibrations, respectively. The band between 1400 cm^{-1} and 1500 cm^{-1}, which corresponds to the $C_a = C_b$ symmetrical vibration on the five-member ring of PEDOT, can be used to distinguish the molecular structure of the PEDOT chains [11, 25, 33, 35, 36]. Conceptually, the benzoid form of PEDOT is a de-doping state while the quinoid form is a doping state, which means that the carrier concentration in the PEDOT chains is proportional to the quinoid form/benzoid form ratio. In other words, the quinoid form of PEDOT (doping state) has a larger work function when compared with the benzoid form (de-doping state).

8.3 Optical and Electrical Characteristics of PEDOT:PSS

For a quinoid form of PEDOT chains in PEDOT:PSS thin films, the electrons are removed from the PEDOT chains due to the strong electron attraction of the sulfonate groups in the PSS segments, which results in the generation of free carriers with lattice distortions (positive polarons). The generation of polarons increases the electrical conductivity, optical transparency in the visible region and reflectance in the infrared region of p-doped PEDOT polymers [37, 38]. In p-doped PEDOT polymers, two polaron states (P1 and P2) are created in the midgap of the π-π^* transition. Figure 8.2a presents the electronic structure of a heavily p-doped PEDOT polymer. Figure 8.2b presents the transmittance spectrum of a PEDOT:PSS thin film (1:6 wt%) on glass substrate. The transmission drop in the violet region is due to the absorption of the glass substrate. The transmittance in the visible region (400 nm to 700 nm) is higher than 88%, which indicates that the exciton transition in the PEDOT chains is nearly completely suppressed due to the generation of polarons. The strong free carrier effect results in a reduced transmittance in the long wavelength region, which indicates that doping of the PEDOT polymer by PSS produces a heavily p-doped organic semiconductor. The resultant broad absorption

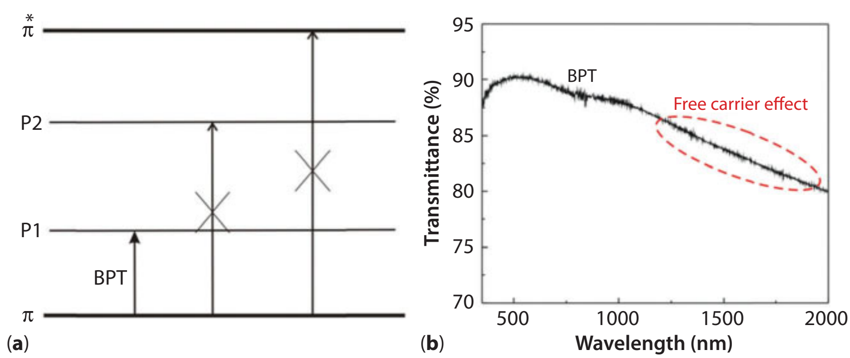

Figure 8.2 (a) Electronic structure of p-doped PEDOTs. (b) Transmittance spectrum of a PEDOT:PSS thin film deposited on glass substrate.

drop (570–1050 nm) can be assigned to the bipolarons transition (BPT) from π to P1. The absorption peak of the PSS polymers is located at ~260 nm [39] while the broad photoluminescence of the PSS polymers ranges from 413 nm (3.0 eV) to 1033 nm (1.2 eV) [38]. The PSS polymer is highly transparent in the near-infrared region, which indicates that the free carrier density of the PSS segments in the PEDOT:PSS thin film is quite low and suggests that they act as electrical insulators when compared to the PEDOT chains. This means that the PEDOT chains in PEDOT:PSS thin films are the hole transport pathways, which is confirmed by the conducting AFM images seen in refs. [40, 41].

The transmittance of the PEDOT:PSS thin film (1:6 wt%) decreases monotonically with the wavelength in the near-infrared region, which is similar to the transmittance spectra of conductive transparent oxide films (Al-doped ZnO, Ga-doped ZnO and ITO) [42–44]. This means that the free carrier density of the PEDOT chains in the PEDOT:PSS thin film is higher than $1\times10^{20}/cm^3$. However, the PEDOT chain connectivity dominates the electrical conductivity of the PEDOT:PSS thin films and impedes the determination of the carrier characteristics (carrier density and carrier mobility) of the PEDOT chains in the films when using Hall measurements. The PEDOT chain connectivity can be evaluated by examining various AFM images (phase, adhesion and current) [11, 41, 45]. Owing to the disconnected PEDOT chains, it is not easy to use electrical measurement methods to investigate the carrier characteristics of the PEDOT chains in PEDOT:PSS thin films. In our previous report [11], the carrier characteristics of the PEDOT chains can be obtained by fitting the near-infrared transmittance spectrum of the PEDOT:PSS thin film with the transfer matrix method and the Drude model. Figure 8.3 presents the transmittance spectra and fitting curves of the PEDOT:PSS thin films on glass substrate [25]. The PEDOT:PSS nanocomposite thin film can be viewed as an effective medium because the PEDOT chains are randomly distributed within the PSS host. In the near-infrared region, the dielectric constant of the PSS segments is assumed to be 2.56 while the dielectric constant of the PEDOT chains can be described by the free carrier model (Drude model) as follows:

$$\varepsilon_{PEDOT} = \varepsilon_{\infty} - \frac{\omega_P^2}{\omega\left(\omega + j\nu_c\right)}, \tag{8.1}$$

where ε_{∞} is assumed to be 2.56 [46]; ω_p is the plasma frequency of the free carriers; ω is the angular frequency of the incident lights; and ν_c is the collision frequency of the free carriers. The free carrier density, carrier

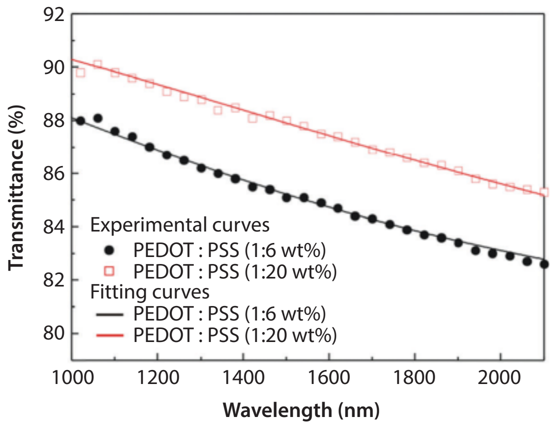

Figure 8.3 Transmittance spectra of the PEDOT:PSS thin films on glass substrate. (Reprinted with permission from [25])

Table 8.1 Free carrier density, carrier mobility and local conductivity of the PEDOT chains in the PEDOT:PSS thin films. (Adapted with permission from [25])

PEDOT/PSS ratio	N_{PEDOT} (1/cm³)	μ_{PEDOT} (cm²/Vs)	σ_{PEDOT} (S/cm)
1/6	1.191×10^{21}	5.32	1014
1/20	1.438×10^{21}	8.36	1923

mobility and local conductivity of the PEDOT chains can be calculated by $N_{PEDOT} = \varepsilon_0 m_h^* \omega_P^2 / e^2$, $\mu_{PEDOT} = e / m_h^* \nu_c$ and $\sigma_{PEDOT} = \varepsilon_0 \omega_P^2 / \nu_c$, respectively, where ε_0 is the absolute permittivity, the effective carrier mass, m_h^*, is assumed to be 0.3 m_e [47]; m_e is the electron mass; and e is the electric charge. The effective dielectric constant of a PEDOT:PSS thin film can be obtained using the Maxwell Garnett model [48]. During the curve fitting process, the dispersive dielectric constant of the glass substrate and the thickness of the PEDOT:PSS thin film have to be known and are fixed, while the plasma frequency and collision frequency are varied. The dielectric constant of the PEDOT chains is adjusted in order to fit the experimental data. After the curve fitting process, the free carrier density, carrier mobility and local conductivity of the PEDOT chains in the PEDOT:PSS thin film are obtained and are listed in Table 8.1. The experimental results show that the local conductivity of the PEDOT:PSS (1:20 wt%) thin film is larger than that of the PEDOT:PSS (1:6 wt%) thin film because of the larger free carrier density and the higher carrier mobility in the PEDOT chains. This result implies the conformation of the PEDOT changes from a benziod-like form (coil conformation, de-doping state) to a quinoid-like form (linear conformation, doping state) with a decrease of the PEDOT/PSS ratio, which is also confirmed by the Raman fingerprints of the PEDOT:PSS thin films at the $C_a = C_b$ stretching mode (~1450 cm^{-1}) in the five-member ring of PEDOT chains.

8.4 Electronic Characteristics of PEDOT:PSS

PEDOT is a conjugated thiophene-derivative polymer, which can be doped to achieve high electrical conductivity. Thiophene-derivatives fall into the class of nondegenerate ground-state polymers [49–51], which results in polaron or bipolaron states for charged defect in the polymer. A polaron is a radical-ion coupled with a local geometry relaxation, which has electron spin resonance characteristics. At higher doping concentrations, the paired polarons form a doubly charged spinless bipolaron which has a strongly localized lattice relaxation. Figure 8.4 presents the electronic structures of the neutral PEDOT, lightly p-doped PEDOT and heavily p-doped PEDOT. The energy difference between the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) of PEDOT polymers can be increased from 1.6 eV [52] to 2.6 eV [37] by doping with PSS polymers, which is the signature of heavily p-doped conducting polymers [37]. The band gap widening effect of the PEDOT polymers is due to the fact that bipolaron states entering into the gap are taken from the HOMO and LUMO edges. In heavily p-doped PEDOT polymers, the conduction mechanism is highly unexpected in the sense that all bands are either completely filled or vacant and that mobile spinless bipolarons, not electrons, transport the current. The Fermi level of the PEDOT:PSS thin films can be

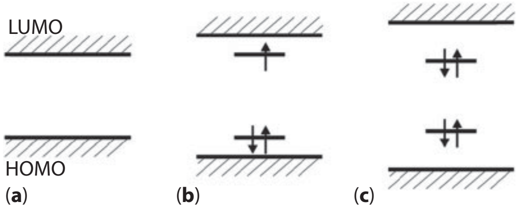

Figure 8.4 Energy level scheme of the self-localized states in PEDOT polymers. (a) Band edge of a neutral PEDOT polymer; (b) polaron state formed upon the addition of an extra electron; (c) spinless bipolaron state formed upon the addition of a second electron.

manipulated to increase from −5.20 eV to −5.44 eV by increasing the PSS/PEDOT ratio from 10 to 46 [40]. For a heavily p-doped PEDOT:PSS thin film, the Fermi level (E_F) is close to the HOMO energy level (E_{HOMO}) [53]. Therefore, the LUMO energy level (E_{LUMO}) of a PEDOT:PSS thin film can be determined by the relation: $\mathrm{E_{LUMO} = E_F + E_G}$ when the optical band gap, E_G, is known.

8.5 Highly Conductive PEDOT:PSS Thin Films

These heavily PSS polymer doped p-type PEDOT polymers can be expected to serve as high-performance TCEs in organic optoelectronic devices due to their high transparency in visible region (400–700 nm) and the high carrier density (~10^{21}/cm^3) of the PEDOT polymers. However, the electrical conductivity of PEDOT:PSS (1:2.5 wt%) thin films is lower than 1 S/cm which is far smaller than that of ITO thin films (~7000 S/cm). The main reason for this is the poor connection between the PEDOT chains in the PEDOT:PSS thin films, as shown in Figure 8.5d, which impedes electrical conductivity. However, the electrical conductivity of the thin films can be increased from 0.84 S/cm to 1210 S/cm using a post-immersion treatment process [11]. There is a gradual increase in the PEDOT chain connectivity with an increase in the immersion time of the PEDOT:PSS thin film in an ethylene glycol/hexafluoro-isopropyl alcohol solvent mixture, as shown in Figure 8.5d–f. The increase in the PEDOT chain connectivity is widely explained as being due to the partial removal of PSS polymers [22], which results in a lot of holes, as shown in Figure 8.5a–c. After the post-treatment process, the free carrier density of the PEDOT polymers increases from 5.0×10^{20}/cm^3 to 8.2×10^{20}/cm^3, which results in an improvement in the local conductivity of the PEDOT polymers from 944 S/cm to 1284 S/cm. This result is due to the conformational change of the PEDOT chains from a benzoid-like form to a quinoid-like form, as confirmed by the Raman fingerprints of the PEDOT:PSS thin films [11, 25, 33, 35, 36].

In previous studies [16–18], it was found that PEDOT:PSS thin-film-based TCEs could be fabricated on top of glass (plastic) substrates using various pretreatment and post-treatment methods. The addition of alcohols to the PEDOT:PSS (1:2.5 wt%) aqueous solutions during pretreatment can increase the electrical conductivity of the PEDOT:PSS (1:2.5 wt%) thin films from 0.2 S/cm^{-1} to 103 S/cm^{-1} due to the preferential solvation with cosolvents [54]. When an ethanol-water pretreated PEDOT:PSS thin film is used as the anode electrode, a moderate power conversion efficiency (PCE) of 2.87% can be achieved in P3HT:PCBM-based solar cells. The

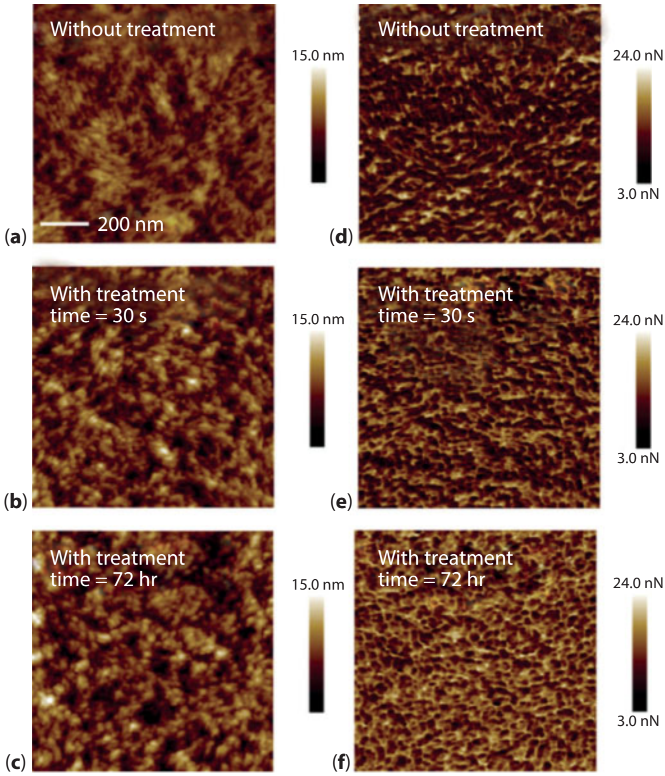

Figure 8.5 Atomic force microscopic images of PEDOT:PSS thin films: (a–c) images of the topography; (d–f) adhesion images. (Reprinted with permission from [11])

electrical conductivity of the PEDOT:PSS (1:2.5 wt%) thin films can be increased from 0.3 S/cm to 1050 S/cm (1100 S/cm) by using a dipping (dropping) method with methanol solvents as part of the post-treatment process [22]. The electrical conductivity of PEDOT:PSS (1:2.5 wt%) thin films can be further improved to 1336 S/cm when a co-treatment (dip and drop) method is used during the post-treatment process [22]. When the methanol post-treated PEDOT:PSS thin film is used as the anode electrode, a high PCE (3.71%) in P3HT:PCBM-based solar cells can be realized. The use of formic acid [12] and H_2SO_4 [55] in the post-treatment process can improve the electrical conductivity of the PEDOT:PSS thin films from 0.3 S/cm to 1900 S/cm and 3065 S/cm, respectively.

The open-circuit voltage (V_{OC}) (0.58 V) in glass/PEDOT:PSS (1:2.5 wt%)/P3HT:PCBM/Ca/Al solar cells is lower than the V_{OC} (0.61V) in glass/ITO/PEDOT:PSS (1:6 wt%)/P3HT:PCBM/Ca/Al solar cells [11] because the work function of the PEDOT:PSS (1:2.5 wt%) thin films is smaller than that of the PEDOT:PSS (1:6 wt%) thin films. However, the work function of modified PEDOT:PSS (1:2.5 wt%) thin films is expected to increase because of increasing the free carrier density of the PEDOT chains.

8.6 Hole-Transporting Materials: PEDOT:PSS Thin Films

The role of the HTM in solar cells is to extract positive carriers (holes) and to block negative carriers (electrons) from the light-absorbing material. Therefore, the HOMO energy level (LUMO energy level) of the HTM needs to be higher than that of the light-absorbing material (active layer), as shown in Figure 8.6. In an ideal solar cell, the V_{OC} can be defined as the energy difference between the LUMO of the electron acceptors and the HOMO of the HTMs. Therefore, the highest theoretical V_{OC} is the optical band gap of active layers when the LUMO (HOMO) of the electron acceptors (HTMs) is equal to the LUMO (HOMO) of the active layers. However, the exciton dissociation and carrier extraction efficiencies at the active layer/HTM (active layer/electron acceptor) interface, which influence the short-circuit current density (J_{SC}), is proportional to the driving electric force which is created due to the energy difference between the HOMO (LUMO) of the active layer and the HOMO (LUMO) of the HTM (electron acceptor). It is clear that there is a trade-off between the V_{OC} and J_{SC} [56], which reduces the maximum efficiency of Shockley-Queisser limit in p-n junction solar cells.

Under a suitable thermal annealing process, the HOMO and LUMO energy levels of the PEDOT:PSS (1:6 wt%) thin films are −5.2 eV and −2.6 eV, respectively, which is an appropriate HTM for solar cells. In

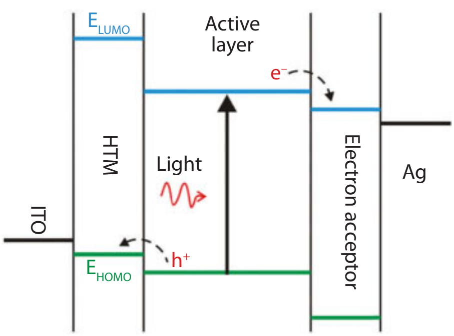

Figure 8.6 Electronic structure of planar solar cells.

addition, the diameter of PEDOT-PSS core-shell clusters are close to the thickness of the PEDOT:PSS thin films, while the PEDOT-PSS core-shell cluster is a PEDOT-rich nanoparticle. This means that the extracted holes can propagate to the anode electrode through the PEDOT-rich nanoparticles. In a previous report [25], we showed the local conductivity of the PEDOT chains in the PEDOT:PSS (1:6 wt%) to be higher than 1000 S/cm, which results in an excellent exciton dissociation (carrier extraction) at the interface between the PEDOT:PSS and the active layer [21]. Therefore, it is possible to increase the V_{OC} without reducing the J_{SC} by lowering (increasing) the LUMO energy level (work function) of PEDOT:PSS thin films. In previous reports [14, 15, 25–28], it was discovered that the work function of the PEDOT:PSS thin films could be increased by varying the PEDOT/PSS ratio, spin rate, thermal annealing condition and viscosity of PEDOT:PSS solutions.

8.6.1 Effect of PEDOT/PSS Ratio

The work function of the PEDOT:PSS thin films can be increased by decreasing the PEDOT/PSS ratio, as shown in Figure 8.7a, which is determined using a photoelectron emission spectrometer. When the PEDOT/PSS ratio decreases from 0.4 to 0.05, the work function (V_{OC}) of the PEDOT:PSS thin films ($CH_3NH_3PbI_3$-based solar cells) increases from 5.02 eV (0.77 V) to 5.23 eV (0.90 V). The increase in the work function (0.21 eV) of the HTM is nearly contributed to the improvement in the V_{OC} (0.23 eV) of the solar cell. Figure 8.7b presents the current density-voltage (J-V) curves of the $CH_3NH_3PbI_3$-based solar cells fabricated using the different PEDOT:PSS

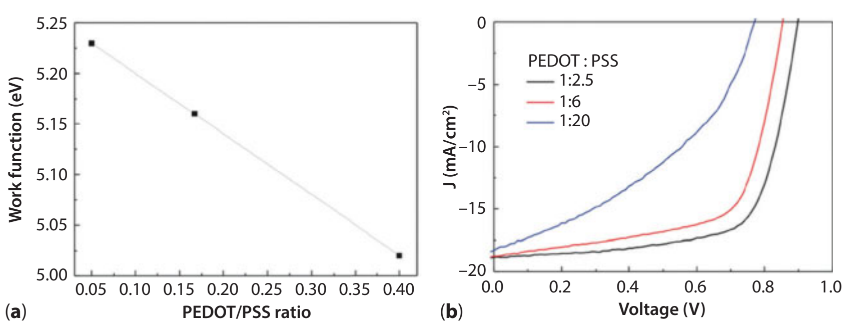

Figure 8.7 (a) Work functions of the PEDOT:PSS thin films. (b) Current density-voltage (J-V) curves of the $CH_3NH_3PbI_3$-based solar cells fabricated with the different PEDOT:PSS thin films under one-sun illumination (AM 1.5G, 100mW/cm^2). (Reprinted with permission from [25])

thin films [25]. The work function of the PEDOT:PSS thin films increases with a decrease in the PEDOT/PSS ratio due to the conformational change of the PEDOT chains from a benzoid-like form to a quinoid-like form. The possible reason for this is that the PEDOT aggregation (benzoid-like form) is suppressed due to the excess PSS polymers and thereby forming linear PEDOT chains (quinoid-like form) doped by PSS segments. However, the excess PSS polymers would degrade the performance of optoelectronic devices due to the low stability of PSS polymers in high-moisture environments [57]. In addition, the value of J_{SC} is not reduced (see Figure 8.7b) when the PEDOT/PSS ratio decreases from 1/6 to 1/20, which originates from the increased local conductivity of the PEDOT polymers (see Table 8.1).

8.6.2 Effect of Spin Rate

The spin rate influences the thickness of the PEDOT:PSS thin films and the evaporation speed of the PEDOT:PSS aqueous solutions during the spin-coating process. Usually, the PEDOT:PSS thin films are deposited on top of ITO electrode as the HTM in solar cells. In order to avoid the electron injection from the active layer to the ITO electrode, the ITO surface has to be completely covered by the PEDOT:PSS thin film which is an electron-blocking layer (EBL). In references [15, 58–60], it can be found that the optimal thickness of PEDOT:PSS (1:6 wt%) thin films is about 40 nm for ITO-based organic (perovskite) solar cells.

The work function of the PEDOT:PSS (1:6 wt%) thin films can be increased from 5.15 eV to 5.39 eV by increasing the spin rate from 3000 rpm to 6000 rpm, which results in a (an) decrease (increase) in the thickness (surface roughness) of the PEDOT:PSS thin films from 55 nm (1.43 nm) to 25 nm (2.19 nm) [15]. Therefore, it is predicted that the V_{OC} of the solar cells can be improved by increasing the spin rate of the PEDOT:PSS aqueous solutions. The photovoltaic performance of the mixed-organic-cation (MOC) perovskite-based solar cells is listed in Table 8.2. The experimental results show that the V_{OC} of the solar cells increases from 0.932 V to 0.979 V with an increase in the spin rate for the fabrication of the PEDOT:PSS thin films, which originates from the increased work function of the PEDOT:PSS thin films from 5.15 eV to 5.20 eV. In addition, the J_{SC} of the MOC perovskite-based solar cells can also be increased from 19.45 mA/cm^2 to 20.96 mA/cm^2 due to the better exciton dissociation efficiency, which is confirmed by a photoluminescence (PL) quenching experiment, as shown in Figure 8.8. In order to avoid the strong light absorption of the PCBM thin films in the violet region, the samples are

Table 8.2 Photovoltaic performance of the mixed-organic-cation perovskite-based solar cells under 1 sun illumination (AM 1.5G, 100mW/cm^2). (Adapted with permission from [15])

Spin rate (rpm)[a]/ TK(nm)[b]/Rq(nm)[c]	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	PCE (%)
3000/55/1.43	0.932 ± 0.025	19.45 ± 0.12	67.4 ± 0.5	12.21 ± 0.15
4000/46/1.66	0.940 ± 0.093	20.09 ± 0.17	65.9 ± 0.6	12.45 ± 0.22
5000/37/1.84	0.979 ± 0.016	20.96 ± 0.14	64.7 ± 1.2	13.28 ± 0.37
6000/25/2.19	0.953 ± 0.010	20.66 ± 0.39	56.3 ± 4.9	11.08 ± 0.83

[a]Spin Rate: spin rate of PEDOT:PSS thin film.
[b]TK: thickness of PEDOT:PSS thin film.
[c]Rq: surface roughness of PEDOT:PS thin film.

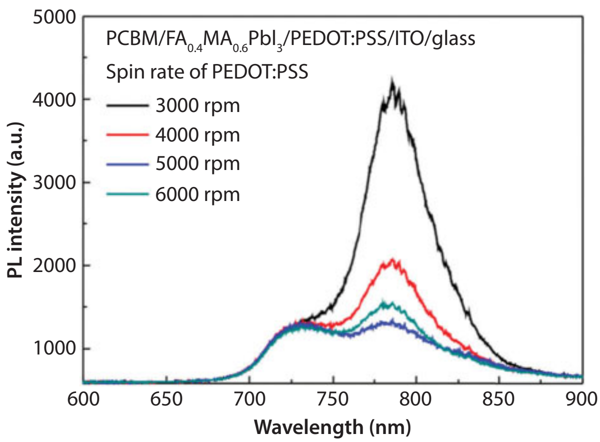

Figure 8.8 Photoluminescence spectra of PCBM/$FA_{0.4}MA_{0.6}PbI_3$/PEDOT:PSS/ITO/glass samples. (Reprinted with permission from [15])

illuminated from the glass side by a 405-nm laser source. The prominent PL peaks at the wavelengths of 720 nm and 770 nm are emitted from the PCBM thin films and the MOC perovskite ($FA_{0.4}MA_{0.6}PbI_3$) thin films, where FA and MA are formamidinium ($HC(NH_2)_2$) and methylammonium (CH_3NH_3), respectively. Half of the photo-generated excitons in perovskite thin films can self-dissociate due to the small exciton binding energy (2 meV ~ 70 meV) [61–66]. The residual excitons must diffuse to the interfaces for the generation of photocurrents. Therefore, better exciton dissociation results in a lower PL intensity which corresponds to the higher J_{SC}.

The V_{OC} and J_{SC} (see Table 8.2) are simultaneously enhanced by increasing the spin rate of the PEDOT:PSS thin films from 3000 rpm to 5000 rpm. However, the fill factor (FF) of the MOC perovskite-based solar cells decreases from 67.4% to 56.3% with an increase in the spin rate of the PEDOT:PSS thin films from 3000 rpm to 6000 rpm. The trend of the FF of the solar cells is proportional to the trend of the thickness of the PEDOT:PSS thin films, which reflects the fact that the surface of the ITO thin films is not completely covered by the thin PEDOT:PSS layers and the contact between the perovskite and the ITO decreases the FF due to the carrier recombination at the perovskite/ITO interface. When the spin rate increases from 5000 rpm to 6000 rpm, the carrier recombination at the perovskite/ITO interface results in the simultaneous reductions in the V_{OC} and J_{SC}, leading to an optimal average PCE of 13.28% for the inverted MOC perovskite-based solar cells.

8.6.3 Effect of Thermal Annealing Temperature

For air-dried PEDOT:PSS thin films [24], the PEDOT chains are randomly distributed in the PEDOT:PSS thin films, which means that there is no clear phase separation between amorphous and crystalline phases. Therefore, the extremely low electrical conductivity in air-dried PEDOT:PSS thin films originates from the homogeneous distribution of insulating PSS polymers which impedes the formation of conductive PEDOT channels. The thermal annealing treatment is an important process used to form the conductive PEDOT-PSS core-shell clusters (PEDOT-rich nanoparticles) embedded in the insulating PSS polymers and serving as the carrier extraction pathways in the HTM of solar cells. The electrical conductivity and work function of PEDOT:PSS thin films can be manipulated by altering the annealing temperature after the spin-coating process [26, 27].

PEDOT:PSS thin films can be fabricated on top of various hydrophilic substrates using solution processes such as spin-coating, spray-coating, and roll-to-roll processed methods. The residual solvent (water) can be removed by thermal drying [67], vacuum drying [47] and microwave-assisted drying methods [68], which strongly influences the optical and electrical properties of the resultant PEDOT:PSS thin films. During the evaporation process of the residual solvent, the PSS segments can be aggregated to form the PSS clusters which results in the separation between the PSS segments and the PEDOT chains. Then, the PSS segments and PEDOT chains are rearranged to form the conductive PEDOT-PSS core-shell clusters which dominate the electrical and electronic properties of PEDOT:PSS thin films.

In a PEDOT:PSS aqueous solution, the PEDOT chains are attracted to the PSS segments, which are dissolved in an aqueous solution. When the water molecules evaporates from the PEDOT:PSS aqueous solution by heating the substrate, the aggregations of PSS segments and PEDOT chains weaken the electrical attraction between the negative PSS segments and the positive PEDOT chains. Therefore, the evaporation rate of water molecules strongly influences the aggregation degrees of PSS segments and PEDOT chains, which determines the electrical and electronic properties of the resultant PEDOT:PSS thin films. Conceptually, the fast evaporation rate of water molecules restrains the aggregations of PSS segments and PEDOT chains, which maintains the quinoid-like form of PEDOT chains doped by PSS segments. This means that the faster evaporation rate of water molecules results in the higher free carrier density of PEDOT chains in p-type PEDOT:PSS thin films. Therefore, it can be predicted that the work function of PEDOT:PSS thin films is proportional to the thermal annealing temperature. However, the PSS degrades via ruptures in the sulfonate group from styrene when the annealing temperature is larger than 260 °C, which results in a significant reduction in the V_{OC} of P3HT-based solar cells due to the reduced work function of the p-type PEDOT:PSS thin films. When the thermal annealing temperature is higher than 260 °C, the reduced work function of p-type PEDOT:PSS thin films originates from the reduction of PSS content in PEDOT:PSS thin films, which decreases the p-type doping concentration in PEDOT chains.

8.6.4 Effects of Viscosity of PEDOT:PSS Solutions

In the previous subsections (8.6.1–8.6.3), the effects of the PEDOT/PSS ratio, spin rate and thermal annealing on the properties of PEDOT:PSS thin films and the corresponding photovoltaic performance of the resultant solar cells were discussed. The main conclusion is that the molecular structure of the PEDOT chains in the PEDOT:PSS thin films can be manipulated by varying the PEDOT/PSS ratio, spin rate and thermal annealing temperature. To explore the dynamics of the formation of PEDOT:PSS thin films, the viscosity of the PEDOT:PSS aqueous solution is varied by adding a series of alcohols (methanol, ethanol and isopropyl alcohol) which have different boiling points and viscosities (see Table 8.3).

It is well known that the synthesis of conductive PEDOT:PSS polymers involves the polymerization of EDOT in an aqueous solution containing

Table 8.3 Properties of the solvent additives and the modified PEDOT:PSS thin films (Adapted with permission from [28]).

Solvent additive	Boiling point (°C)	Viscosity (cp)	Surface roughness (nm)	Contact angle (°)	Work function (eV)
Methanol	64.7	0.545	1.53	42	5.06
Ethanol	78.4	1.074	1.47	58	5.10
IPA	82.6	1.960	1.76	49	5.12

long PSS segments, with sodium perfulfate as the oxidizing agent. It is indicated that the conductive PEDOT:PSS polymers in the aqueous solution have a linear molecular form. The fabrication process of modified PEDOT:PSS thin films is described below. Methanol, ethanol and isopropyl alcohol (IPA) are individually added to a PEDOT:PSS (1:6 wt%) aqueous solution, and the volume ratio of water to alcohol is 1:2. The modified PEDOT:PSS aqueous solutions are spin-coated onto the top of the ultraviolet-ozone treated glass substrates, then subjected to thermal annealing at 150 °C for 20 min. After that, the modified PEDOT:PSS/glass samples are cooled on a steel plate at room temperature.

The surface roughness of the PEDOT:PSS thin films is measured using a contact-mode atomic-force microscope. The water droplet contact angle of the PEDOT:PSS thin films is determined by a homemade optical microscope imaging system. The surface morphology and water droplet contact angle of the modified PEDOT:PSS thin films are shown in Figure 8.9. The use of high viscosity (boiling point) solvents for thin-film deposition by a spin-coating method results in a large (small) surface roughness [69]. This means that the surface roughness of the modified PEDOT:PSS thin films is related to the viscosity and boiling point of the solvent additives. The surface roughness and water droplet contact angle of the modified PEDOT:PSS thin films are listed in Table 8.3. The trend of the water droplet contact angle on the surface of the modified PEDOT:PSS thin films is inversely proportional to the trend of the surface roughness of the modified PEDOT:PSS thin films, which implies that a flat PEDOT:PSS thin film has a low surface free energy. The change in the surface free energy of the modified PEDOT:PSS thin films indicates that the arrangement of the structure of the PEDOT chains and PSS segments has changed, which implies that the electronic properties of the PEDOT:PSS thin films can be manipulated by the addition of solvents. To investigate the electronic properties of the modified PEDOT:PSS thin films, a photoelectron emission spectrometer is used to measure the work function (see Table 8.3). The work function of the

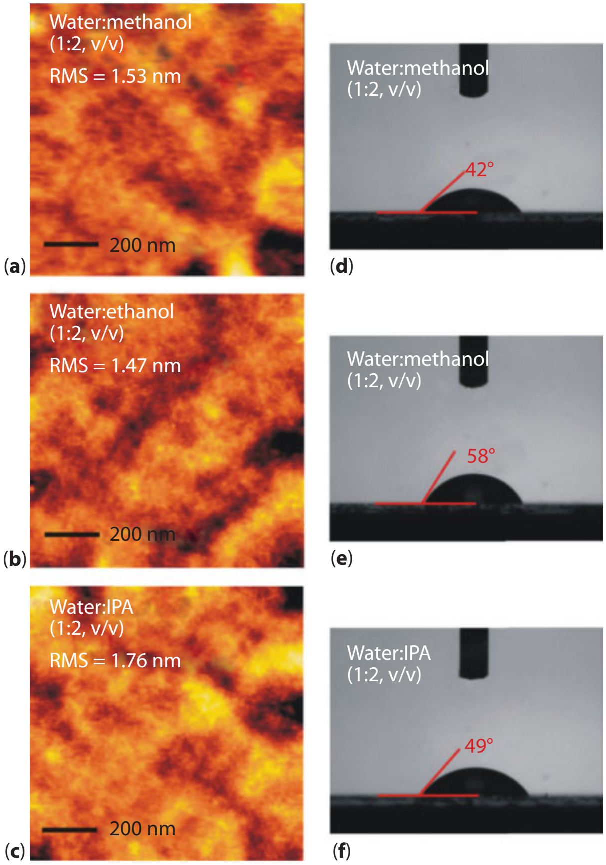

Figure 8.9 Atomic-force microscopic images (a,b,c) and water droplet contact angle images (d,e,f) of the modified PEDOT:PSS thin films (Reprinted with permission from [28]).

modified films is proportional to the viscosity of the solvent additives. The increase in the work function is due to the conformational change of the PEDOT chains from a benzoid-like form to a quinoid-like form, which can be confirmed using the Raman scattering spectra [11, 25, 33, 35, 36].

The experimental results show that the modification of PEDOT:PSS thin film leads to a larger work function and high local conductivity, which is predicted to be advantageous to the V_{OC} and J_{SC} of the resultant solar cells. The higher work function and high local conductivity are produced through the addition of an IPA solvent which increases the viscosity of the PEDOT:PSS solutions and thereby suppresses the aggregation of PSS segments and PEDOT chains during the thin-film deposition process to maintain the linearity of the PEDOT chains.

8.7 Directions for Future Development

The PEDOT:PSS thin films can be used as the TCE or HTM in solar cells due to their high transparency in the visible region, high electrical conductivity and large work function. The transmittance of the PEDOT:PSS thin films can be larger than 90% in the visible region due to the formation of bipolarons, which effectively suppresses the π-π^* transition of the PEDOT polymers. The electrical conductivity can be close to the local conductivity due to the formation of conductive PEDOT channels in the PEDOT:PSS thin film. The work function of the p-doped PEDOT polymers can be adjusted by tuning the molecular structure of the PEDOT chains in the thin films. As a consequence, it is clear that the structural, optical, electrical and electronic properties of the PEDOT:PSS thin films can be arbitrarily manipulated by tailing the molecular structures of PEDOT:PSS nanocomplexes.

The conductivity ($\sigma = eN\mu$) of materials is proportional to the carrier concentration (N) and carrier mobility (μ) in materials, where e is the electric charge. To optimize the intrinsic (local) conductivity of the PEDOT chains in PEDOT:PSS thin films, the carrier concentration and carrier mobility must be simultaneously increased. One of the ways to do this is to change the molecular structure of the PEDOT chains from a benzoid form (de-doping state) to a quinoid form (doping state) by decreasing the PEDOT/PSS ratio (see Table 8.2). The theoretical carrier concentration in heavily p-doped PEDOT chains can be calculated as follows:

$$N_{PEDOT} = \frac{1}{V_{PEDOT}}, \tag{8.2}$$

where V_{PEDOT} (= a×b×c = 0.78 nm × 1.18 nm × 0.69 nm) [70] is the volume of a EDOT unit in the PEDOT chain. Therefore, the theoretical carrier concentration in a p-doped PEDOT chain (quinoid form) can be estimated to be $1.575 \times 10^{21}/\text{cm}^3$. The theoretical carrier concentration in the PEDOT chain is slightly larger than the experimental value (see Table 8.2), which implies that each EDOT unit in the PEDOT:PSS (1:20 wt%) thin film is almost doped by the PSS polymer. So far, the highest electrical conductivity of the modified PEDOT:PSS (1:2.5 wt%) thin film has been 3300 S/cm [71], which was measured by the van der Pauw four-point probe technique. However, the carrier concentration and carrier mobility of the modified PEDOT:PSS (1:2.5 wt%) thin film are not provided in ref. [71]. To the best of our knowledge, the highest reported carrier mobility in PEDOT:PSS thin films is 53.8 cm^2/Vs [72]. Therefore, the optimal electrical conductivity of a PEDOT:PSS thin film is predicted to be larger than 13557 S/cm (= 1.6×10^{-19}C $\times$ 1.575×10^{21}cm^{3} $\times$ 53.8 cm^{2}/Vs), to be used as a standalone electrode.

However, it is known that PEDOT:PSS thin films are unstable under humid environments [73] or high temperature treatments [9, 73]. To improve the stability, a sulfonated poly(imide) (SPI) polymer can be substituted for the PSS polymer. However, the doping concentration of a PEDOT:SPI thin film is lower than that of the PEDOT:PSS thin films due to fewer doping sites. The carrier concentration in the PEDOT chains can be increased to 10^{23}/cm^{3} so that when the conductive PEDOT-based polymer is synthesized using iron trifluoromethanesulfonate as the oxidant it exhibits metallic behaviors [74]. The results show that the structure, optical, electrical and electronic properties of stable conductive PEDOT-based polymers can be adjusted to obtain the desired characteristics by using different chemical dopants.

8.8 Conclusion

A comprehensive study has been carried out to explore the structural, optical, electrical and electronic properties of PEDOT:PSS thin-film-based conductive transparent electrodes and hole-transporting materials under various fabrication conditions. The formation of bipolarons in the heavily p-doped PEDOT polymers, which dominates the optical, electrical and electronic properties of the PEDOT:PSS thin films, results in high transparency in the visible region (suppressed exciton transition), a transmission drop centered at the wavelength of 810 nm (π to P1 transition), and a reduction in the transmittance in the long wavelength region (free carrier effect).

For the PEDOT:PSS (1:2.5 wt%) thin-film-based conductive transparent electrode, the removal of the excess PSS polymers greatly increases the electrical conductivity from ~1 S/cm to ~1000 S/cm due to the formation of a 3D PEDOT network and an increase in the local conductivity in the PEDOT chains. Moreover, the highest electrical conductivity of PEDOT:PSS thin films is predicted to be larger than 13557 S/cm, which is higher than the highest reported experimental value (3300 S/cm). Therefore, there is still room for improvement of the electrical conductivity of PEDOT:PSS thin films.

For the PEDOT:PSS thin-film-based HTMs, the work function, which is related to the molecular structure of the PEDOT chains, can be manipulated by varying the PEDOT/PSS ratio, spin rate, thermal annealing temperature and viscosity of PEDOT:PSS solutions. The work function of the linear PEDOT chains (quinoid form, doping state) is larger than that of the coiled PEDOT chains (benzoid form, de-doping state). Owing to the high exciton dissociation (carrier extraction) at the PEDOT/light-absorbing material interface, the open-circuit voltage of solar cells can be increased without reducing the short-circuit current density of solar cells when increasing the work function of the PEDOT:PSS thin film.

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